Miisky Project — Workflow Overview

A nutrition & diet management platform connecting **Patients**, **Nutritionists**, **Micro Kitchens**, **Supply Chain**, and **Food Buyers**.

Tech Stack

Layer	Technology
Backend	
Framework	Django
API Layer	Django REST Framework
Language	Python
Database	PostgreSQL
Frontend	
Library	React.js

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1. User Roles & Access Flow

```
┌──────────────────────────────────────────────────────────────────────────┐
│                           USER REGISTRATION & AUTH                        │
├──────────────────────────────────────────────────────────────────────────┤
│  Roles: admin | nutritionist | patient | supply_chain | food_buyer |      │
│        micro_kitchen | non_patient                                        │
│                                                                           │
│  Flow: Register → Login (JWT) → Role-based access                         │
└──────────────────────────────────────────────────────────────────────────┘
```

Role	Primary Functions
Admin	User management, food system, health parameters, diet plans
Nutritionist	Patient mapping, health report review, diet suggestions, meetings
Patient	Questionnaire, health reports, diet plans, meals, meetings (with nutrition consultation)
Non Patient	Order food directly from micro kitchen — no nutrition consultation; supply chain delivers
Micro Kitchen	Kitchen profile, inspections, meal prep, marks food ready for delivery
Supply Chain	Pick up from kitchen, deliver to patient / non patient
Food Buyer	Food procurement

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2. Core Workflows

2.1 Patient ↔ Nutritionist Flow

```
Patient                    Nutritionist                 System
   │                            │                          │
   ├── Register & Questionnaire ──────────────────────────►│
   │                            │                          │
   ├── Upload Health Reports ─────────────────────────────►│ PatientHealthReport
   │                            │                          │
   │                            ├── Review Reports ────────►│ NutritionistReview
   │                            │                          │
   │                            ├── Assign Nutritionist ───►│ UserNutritionistMapping
   │                            │   (is_patient_mapped)    │
   │                            │                          │
   │                            ├── Suggest Diet Plan ─────►│ UserDietPlan (suggested)
   │                            │                          │
   ├── Approve/Reject Plan ◄────────────────────────────────│
   │   (approve → payment_pending)                         │
   │                            │                          │
   ├── Pay ◄──────────────────────────────────────────────│ UserDietPlan (active)
   │                            │                          │
   │                            ├── Allot Meals ───────────►│ UserMeal (per day)
   │                            │                          │
   ├── Request Meeting ───────────────────────────────────►│ MeetingRequest
   │                            │                          │
   │                            ├── Approve/Reject ────────►│ meeting_link, scheduled_datetime
   │                            │                          │
   └── Attend Meeting                                        │
```

2.2 Diet Plan Lifecycle (UserDietPlan)

```
suggested → approved → payment_pending → active → completed
                 │
                 └── rejected
```

Status	Description
`suggested`	Nutritionist suggested based on review
`approved`	Patient accepted (before payment)
`rejected`	Patient declined
`payment_pending`	Awaiting payment
`active`	Plan running (meals allotted)
`completed`	Plan period ended
`stopped`	Manually stopped

2.3 Micro Kitchen Flow

```
Micro Kitchen                    Admin/Inspector             System
       │                              │                        │
       ├── Create Profile ──────────────────────────────────────►│ MicroKitchenProfile (draft)
       │   (FSSAI, equipment, cuisine, photos)                  │
       │                              │                        │
       │                              ├── Schedule Inspection ─►│ MicroKitchenInspection
       │                              │   (12 rating criteria)  │
       │                              │                        │
       │                              ├── Submit → Approved ───►│ MicroKitchenProfile
(approved)
       │                              │   or Rejected          │
       │                              │                        │
       └── Kitchen ready for meal orders                        │
```

Inspection Status: `draft` → `submitted` → `approved` / `rejected`

Kitchen Profile Status: `draft` → `approved` / `rejected`

2.4 Micro Kitchen Assignment & Food Delivery Flow

```
Nutritionist / Patient          Micro Kitchen              Supply Chain              Patient
        │                            │                          │                      │
        ├── Allot kitchen to patient ─►│                          │                      │
        │   (or patient selects)     │                          │                      │
        │                            │                          │                      │
        │                            ├── Prep meals as per plan   │                      │
        │                            │                          │                      │
        │                            ├── Mark "Food Ready" ──────►│                      │
        │                            │                          │                      │
        │                            │                          ├── Pick up from kitchen
        │                            │                          │                      │
        │                            │                          ├── Deliver ────────────►│
        │                            │                          │                      │
        └────────────────────────────────────────────────────────┴──────────────────────┘
```

Step	Who	Action
1	Nutritionist **or** Patient	Assign/select micro kitchen for the patient
2	Micro Kitchen	Prepare meals as per allotted plan
3	Micro Kitchen	Mark food ready for supply chain
4	Supply Chain	Pick up from kitchen, deliver to patient
5	Patient	Receive delivery

2.5 Non-Patient Order Flow (No Nutrition Consultation)

Non patients order food directly from micro kitchen, without diet plan or nutritionist.

```
Non Patient                  Micro Kitchen              Supply Chain
     │                            │                          │
```

```
    ├── Browse kitchens ──────────►│                          │
    │                              │                          │
    ├── Order food (direct) ──────►│                          │
    │                              │                          │
    │                              ├── Prepare order          │
    │                              │                          │
    │                              ├── Mark "Food Ready" ────►│
    │                              │                          │
    │                              │                          ├── Pick up
    │                              │                          │
    │                              │                          ├── Deliver ────► Non Patient
    │                              │                          │
    └──────────────────────────────────────────────────────────┘
```

Step	Who	Action
1	Non Patient	Browse micro kitchens, place order (no diet plan)
2	Micro Kitchen	Prepare order, mark food ready
3	Supply Chain	Pick up, deliver to non patient

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3. Food & Nutrition Data Flow

3.1 Recipe / Food Structure

```
MealType (Breakfast, Lunch, etc.)
    │
    └── Food (Idli, Chapati, etc.)
            ├── FoodNutrition (calories, protein, etc.)
            ├── FoodIngredient ──► Ingredient + Unit + quantity
            └── FoodStep (cooking instructions)
```

3.2 Indian Food Composition (FoodName-based)

```
FoodGroup (e.g. CEREALS, FRUITS)
    │
    └── FoodName
            ├── FoodProximate (protein, fat, fiber, energy)
            ├── FoodWaterSolubleVitamins (B1, B2, B3, B6, B9, C)
            ├── FoodFatSolubleVitamins (retinol, tocopherols)
            ├── FoodCarotenoids
            ├── FoodMinerals
            ├── FoodSugars
            ├── FoodAminoAcids
            ├── FoodOrganicAcids
            ├── FoodPolyphenols
            ├── FoodPhytochemicals
            └── FoodFattyAcidProfile
```

3.3 Meal Allotment

```
UserDietPlan (active)
    │
    └── UserMeal (per day)
            ├── meal_type (Breakfast/Lunch/Dinner)
            ├── food (from Food)
            ├── quantity
            ├── meal_date
            └── is_consumed, consumed_at
```

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4. Health & Lab Parameters

```
HealthParameter (e.g. Hemoglobin, Glucose)
    │
    └── NormalRangeForHealthParameter
            ├── min_value, max_value
            ├── unit
            ├── qualitative_value
            └── reference_text
```

- Patient uploads **PatientHealthReport** (blood test, scan, etc.)
- Nutritionist creates **NutritionistReview** with comments
- **UserQuestionnaire** stores diet pattern, allergies, health conditions, etc.

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5. Diet Plans & Features

```
DietPlans (templates)
    ├── title, code, amount, discount_amount, no_of_days
    └── DietPlanFeature (e.g. "Personalized diet chart", "24/7 support")
```

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6. Location Hierarchy

```
Country → State → City
```

Used by: `UserRegister`, `MicroKitchenProfile` (lat/long for maps)

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7. Menu & Functionality by Role

Role	Menu	Submenu	Functionality
admin	Dashboard	—	Main overview of the platform
admin	Locations	Countries	Add, edit, manage country list
admin	Locations	States	Manage states linked to countries

admin	Locations	Cities	Manage cities linked to states
admin	Food Management	Meal Type	Define meal types (Breakfast, Lunch, Dinner, Snacks)
admin	Food Management	Cuisine Type	Define cuisine types (North Indian, South Indian, etc.)
admin	Food Management	Foods	Add, edit food items with meal & cuisine types
admin	Food Management	Units	Define measurement units (Gram, Cup, etc.)
admin	Food Management	Ingredients	Manage raw ingredients used in recipes
admin	Food Management	Recipe Management	Create recipes: link food → ingredients + cooking steps
admin	Food Composition	Food Groups	Group foods (Cereals, Fruits, Oils, etc.)
admin	Food Composition	Food Names	Add food names under groups for nutrition database
admin	Food Composition	Proximate	Store protein, fat, fiber, energy per food
admin	Food Composition	Water Soluble Vitamins	B1, B2, B3, B6, B9, C values per food
admin	Food Composition	Fat Soluble Vitamins	Retinol, tocopherols per food
admin	Food Composition	Carotenoids	Lutein, lycopene, beta-carotene per food
admin	Food Composition	Minerals	Calcium, iron, zinc, etc. per food
admin	Food Composition	Sugars	Starch, fructose, glucose per food
admin	Food Composition	Amino Acids	Amino acid profile per food
admin	Food Composition	Organic Acids	Oxalate, citric acid, etc. per food
admin	Food Composition	Polyphenols	Phenolic acids per food
admin	Food Composition	Phytochemicals	Oligosaccharides, phytosterols per food

admin	Food Composition	Fatty Acid Profile	SFA, MUFA, PUFA per food
admin	User Management	—	Create, edit users (all roles)
admin	Micro Kitchens	Kitchen Information	View kitchen profiles & inspection reports
admin	Health Monitoring	Health Parameters	Define lab parameters (Hemoglobin, Glucose, etc.)
admin	Health Monitoring	Normal Ranges	Set min/max normal values per parameter
admin	Health Monitoring	Diet Plans	Create diet plan templates with features & pricing
admin	Mappings	User–Nutritionist Mapping	Map patients to nutritionists
patient	Questionnaire	—	Fill diet, health, allergies, preferences
patient	Nutritionist Allotted	—	See assigned nutritionist
patient	Diet Plans	—	View active diet plans
patient	Suggested Plans	—	View & approve/reject plans suggested by nutritionist
patient / non_patient	Micro-Kitchens	—	Browse approved kitchens; patient: select or view allotted kitchen; non_patient: order food directly (no consultation)
patient	Health Reports	—	Upload blood tests, scans, prescriptions
patient	Meals Allotted	—	View daily meal plan & mark consumed
patient	Consultation	—	Request meeting with nutritionist
nutritionist	Questionnaire	—	Fill nutritionist profile / settings
nutritionist	Allotted Patients	—	View patients assigned to them
nutritionist	Meal Optimizer	—	Allot meals per patient per day; allot micro kitchen to patient
nutritionist	Patient Documents	—	Review uploaded reports & add comments
nutritionist	Suggest Plan	—	Suggest diet plan to patient
nutritionist	Approved Plans	—	View approved & active plans

nutritionist	Meeting Requests	—	Approve or reject patient meeting requests
micro_kitchen	Questionnaire	—	Fill kitchen profile (FSSAI, equipment, photos)
micro_kitchen	Inspection Report	—	View inspection status & ratings; mark food ready for supply chain
supply_chain	Delivery Questionnaire	—	Fill vehicle, license, insurance; pick up from kitchen & deliver to patient

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8. Key Relationships Diagram

```
                    UserRegister
                         │
    ┌────────────────────┼────────────────────┐
    │                    │                    │
    ▼                    ▼                    ▼
NutritionistProfile  MicroKitchenProfile   DeliveryProfile
    │                    │
    │                    ▼
    │              MicroKitchenInspection
    │
    ├── UserNutritionistMapping ◄── Patient
    │
    ├── NutritionistReview ◄── PatientHealthReport
    │
    ├── UserDietPlan ◄── DietPlans
    │        │
    │        ├── UserMeal ◄── Food, MealType
    │        │
    │        └── MeetingRequest
    │
    └── UserQuestionnaire
```

References: `backend/app/models.py` , `frontend/src/layout/MasterLayout/MasterSidebar.tsx`